

Drug Summary

Everolimus (RAD001) is an oral inhibitor of the mTOR kinase, used in advanced breast cancer to overcome resistance in certain subtypes. It is a **rapamycin analog (rapalog)** that selectively inhibits the mTORC1 complex, thereby blocking downstream signaling in the PI3K/AKT/mTOR pathway ¹ ². Everolimus was FDA-approved (2012) for **hormone receptor-positive, HER2-negative (HR+/HER2-) metastatic breast cancer** in combination with endocrine therapy after prior aromatase inhibitor failure ³ ⁴. In this luminal subtype, everolimus significantly improved progression-free survival (PFS) when added to exemestane, restoring sensitivity in tumors that had become resistant to endocrine therapy ³ ⁵. Everolimus has also been evaluated in HER2-positive disease and triple-negative breast cancer (TNBC). In HER2+ metastatic breast cancer, adding everolimus to trastuzumab-based therapy did not improve outcomes overall, but **subgroup analysis showed a PFS benefit in HER2+ tumors lacking hormone receptors** ⁶. In **TNBC**, everolimus is not standard of care, but early trials suggest it may enhance chemotherapy efficacy in PI3K-hyperactivated cases. A 2025 phase II study (everolimus + carboplatin vs carboplatin) in advanced TNBC showed a 52% reduction in risk of progression or death (HR=0.49) with the combination ⁷ ⁸, reflecting the high frequency of PTEN loss and mTOR pathway activation in TNBC ⁹. Overall, everolimus's role in breast cancer is as a targeted therapy **across molecular subtypes** where tumor growth is driven by dysregulated PI3K/mTOR signaling – most prominently in endocrine-resistant HR+ disease, with ongoing exploration in HER2+ and triple-negative tumors.

Identifiers & Synonyms

- **ChEMBL ID:** ChEMBL1908360 ¹⁰
- **DrugBank ID:** DB01590 ¹¹
- **Synonyms/Trade Names:** Everolimus is also known as **40-O-(2-hydroxyethyl)-rapamycin** (42-O-(2-hydroxy)ethyl rapamycin) ¹². Brand names include **Afinitor** (oncology use) ¹² and **Afinitor Disperz, Zortress** (transplant indication) ¹², **Votubia** (EU brand) ¹³, and **Certican** (ex-U.S. transplant brand) ¹⁴. Development codes for everolimus include **RAD001** (RAD-001) ¹⁵ and **RAD 666** ¹⁶. (Everolimus is an analog of sirolimus/rapamycin, to which it is structurally related ¹⁷.)

Mechanism of Action (Breast Cancer Context)

Everolimus exerts antitumor effects by **allosterically inhibiting the mTORC1 complex**, a master regulator of cell growth, metabolism, and survival. In cells, everolimus binds to the intracellular immunophilin **FKBP12**, forming a drug-protein complex that docks on mTORC1 and prevents mTOR activation ². This **selectively inhibits mTORC1 kinase activity** (and does not significantly inhibit mTORC2) ¹, leading to blockade of downstream signaling – notably reducing phosphorylation of S6K1 and 4E-BP1 – which in turn **suppresses protein synthesis, cell-cycle progression, and cell proliferation** ¹⁸. In breast cancer, everolimus's MOA is context-dependent: in HR+ tumors, the PI3K/AKT/mTOR pathway is frequently hyperactivated as an adaptive escape from hormone dependence ¹⁹ ⁴. By inhibiting mTORC1, everolimus **interrupts this survival pathway** and restores sensitivity to endocrine therapy. Indeed, everolimus can **downregulate estrogen receptor (ER) signaling** in ER+ breast cancer cells – treatment led

to marked reduction of ER mRNA/protein and transcriptional activity in AI-resistant cell lines ²⁰. This diminishes ER-driven proliferation and helps overcome hormonal resistance. Everolimus also induces G1 cell-cycle arrest via **cyclin D1 downregulation** ²¹ and promotes cellular senescence or apoptosis in some contexts by tipping the balance toward pro-apoptotic factors. Beyond HR+ disease, many breast tumors (including subsets of HER2+ and TNBC) exhibit aberrant PI3K/mTOR signaling (e.g. due to **PIK3CA mutations or PTEN loss**), making them inherently reliant on mTOR-mediated growth signals ⁹. In such tumors, everolimus can produce antiproliferative effects by **starving the cancer cells of oncogenic mTOR signals**, even though it is largely cytostatic (preventing growth) rather than strongly cytotoxic ¹⁸. Notably, mTOR also influences tumor angiogenesis and the immune microenvironment: everolimus's inhibition of HIF-1α/VEGF reduces neovascularization ²² ²³, and as an immunosuppressant it can **impair T-lymphocyte activation** ². These ancillary effects may contribute to its anti-tumor activity (anti-angiogenic) but also caution that everolimus might dampen anti-tumor immune responses. Overall, by **blocking a central growth and survival hub (mTORC1)**, everolimus can arrest tumor progression and resensitize tumors to other therapies, especially in breast cancers driven by PI3K/mTOR pathway dysregulation.

Primary Targets (Human)

- **MTOR (Mechanistic Target of Rapamycin)** – Serine/threonine kinase that forms the catalytic core of mTORC1. Everolimus (as a rapamycin analog) directly targets mTORC1 via FKBP12, thereby inhibiting mTOR's kinase activity ². mTOR (gene **MTOR**, HGNC:3942) is the principal **validated target** of everolimus, mediating its anticancer effects by curtailing the PI3K/AKT/mTOR signaling cascade ¹.
- **FKBP1A (FKBP12)** – FK506-binding protein 12, an intracellular immunophilin. Everolimus first binds to FKBP12; the drug-FKBP12 complex then binds mTORC1's FKBP12-rapamycin binding domain to inhibit mTOR ². While FKBP12 itself is not a cancer-driving enzyme, it is the **direct binding target** of everolimus and required for mTORC1 inhibition. (Note: Everolimus does *not* directly bind the mTOR active site, but through FKBP12 it allosterically inactivates mTORC1 ².)

(No other primary targets are known; everolimus is relatively specific for mTORC1. It does not inhibit mTORC2, and has minimal off-target kinase activity in human cells ¹.)

Pathways (Overview)

Everolimus perturbs multiple key pathways in breast cancer. Below is an overview of the **major pathways influenced by everolimus**, identified by standard gene set names (MSigDB Hallmark, Reactome, KEGG, GO, etc.), along with their relevance to everolimus's mechanism, sensitivity, or resistance:

- **HALLMARK_PI3K_AKT_MTOR_SIGNALING** – *PI3K/AKT/mTOR pathway*. This is the core oncogenic pathway that everolimus targets. Hyperactivation of PI3K-AKT signaling (via PIK3CA mutation, AKT activation, PTEN loss) drives proliferation and endocrine therapy resistance ¹⁹ ⁴. Everolimus inhibits mTORC1 within this cascade, thereby **downregulating PI3K/AKT/mTOR signaling** and halting tumor cell growth (mechanistic *and* sensitivity pathway). In breast cancers addicted to PI3K/mTOR, upregulation of this pathway predicts greater everolimus sensitivity ⁹ ²⁴, whereas re-activation of AKT (feedback loop) can mediate drug resistance ²⁵.

- **HALLMARK_MTORC1_SIGNALING** – *mTORC1 downstream output*. Everolimus directly **suppresses mTORC1 signaling**, which includes protein translation and ribosomal biogenesis. Inhibition of mTORC1 leads to reduced phosphorylation of S6 ribosomal protein and 4EBP1, causing translational repression and cell-cycle arrest ²⁶. This pathway's downregulation is a **mechanistic hallmark** of everolimus action (intimately tied to the above PI3K-AKT-MTOR set). Persistent mTORC1 activation despite drug (e.g. via alternate inputs or mTOR mutation) would confer resistance.
- **HALLMARK_ESTROGEN_RESPONSE** (e.g. Estrogen Response Early/Late) – *Estrogen receptor (ER) signaling pathway*. In HR+ breast cancer, ER-driven transcription promotes tumor growth. Everolimus can modulate this pathway: treatment in ER+ models **dramatically reduced ER expression and activity** ²⁰, likely by disrupting mTOR-dependent ER translation and HSP90 chaperoning. Downregulation of the estrogen response pathway is associated with restored endocrine sensitivity (everolimus makes ER+ tumors more responsive to anti-estrogens) ²⁷. This pathway is subtype-specific (luminal/Hormone-Receptor-positive) and its suppression by everolimus is a **key synergy mechanism** in ER+ breast cancer. Conversely, loss of ER (e.g. ER-negative phenotype) can render endocrine-based therapy with everolimus ineffective, which is a consideration in resistance (if an initially ER+ tumor downregulates ER, it may evade the combined effect).
- **HALLMARK_E2F_TARGETS** (and **G2M_CHECKPOINT**) – *Cell cycle progression gene sets*. mTORC1 activity promotes cell-cycle progression (e.g. via Cyclin D1, c-Myc). Everolimus **downregulates cell-cycle drivers** and induces G1 arrest ²⁶. Genes controlled by E2F (Cyclins, CDKs, etc.) and the G₂/M checkpoint are repressed as cells cease proliferation. This pathway downregulation correlates with everolimus's antiproliferative effect (mechanistic). Tumors that cannot undergo cell-cycle arrest (e.g. due to p53/Rb pathway defects) might exhibit less sensitivity.
- **GO:0006914 – Autophagy** – *Autophagy (self-digestion) pathway*. mTOR normally inhibits autophagy; thus everolimus **upregulates autophagy** in tumor cells ²⁸. While autophagy can be cytotoxic in some settings, in breast cancer it often serves as a survival adaptation. Everolimus-induced autophagy enables cancer cells to recycle nutrients under therapeutic stress, contributing to drug resistance ²⁸. High autophagy gene signature upregulation upon treatment is associated with **reduced sensitivity** (cells escape death by autophagic survival). Combination strategies inhibiting autophagy could therefore enhance everolimus efficacy.
- **HALLMARK_MYC_TARGETS_V1/V2** – *MYC oncogene signaling*. Everolimus, paradoxically, can lead to **upregulation of MYC** in breast cancer cells. Acquired everolimus resistance in ER+ models is driven by epigenetic upregulation of c-Myc, with resistant cells showing enriched MYC gene signatures ²⁹. MYC acts as a bypass oncogene, sustaining proliferation despite mTORC1 blockade. Upregulation of MYC pathway is thus a **resistance mechanism**: silencing MYC resensitizes cells to everolimus ³⁰. Clinically, tumors with high MYC activity (or genomic MYC amplification) may respond poorly to everolimus. Combinations of everolimus with BRD4 inhibitors (to suppress MYC transcription) have shown synergy in preclinical models ³¹.
- **REACTOME_SIGNALING_BY_AKT** (AKT signaling feedback) – mTORC1 inhibition can trigger **feedback activation of AKT and upstream receptor tyrosine kinases**. This is because mTORC1 normally exerts negative feedback on PI3K/AKT via S6K-IRS1; everolimus relieves that feedback, causing a rebound in AKT phosphorylation ³². Upregulation of the AKT pathway (via IGF-1R, HER2, or other RTKs) is associated with **drug resistance**: it reactivates a parallel survival route that bypasses the

mTORC1 block. For example, everolimus-resistant breast cancer cells often exhibit increased AKT activation or PIK3CA mutations that drive pathway reactivation ²⁵. Therapies targeting upstream RTKs or AKT itself may overcome this resistance mechanism.

- **HALLMARK_HYPOXIA** and **VEGF_SIGNALING** – *HIF-1α/VEGF-driven angiogenesis*. mTORC1 activity increases HIF-1α translation, promoting angiogenic factor (VEGF) production. Everolimus **downregulates the STAT3-HIF-1α-VEGF axis**, thereby inhibiting tumor angiogenesis ²² ²³. In breast cancer, this manifests as reduced microvessel density and can potentiate anti-tumor effects (by starving the tumor of nutrients). Thus, angiogenesis pathways are *down-regulated* (a beneficial mechanistic effect). Tumors that find alternative ways to induce angiogenesis (e.g. via FGFs or angiopoietins) could resist therapy. Notably, everolimus's anti-angiogenic property is leveraged in other cancers and likely contributes modestly in breast tumors that are highly vascularized.
- **Immune Checkpoints / T-Cell Activation (ImmuneDB signatures)** – *Immune microenvironment pathways*. mTOR is crucial for T-cell growth and function; everolimus's immunosuppressive action **inhibits IL-2/IL-15-driven T-cell activation and proliferation** ³³. In breast tumors, especially TNBC which may rely on immune surveillance, everolimus might reduce effector T-cell infiltration and activity. Preclinical TNBC studies found everolimus increased PD-L1 expression on tumor vasculature and decreased CD8⁺ T-cell infiltration ³⁴ ³⁵. Therefore, upregulation of immunosuppressive pathways (PD-1/PD-L1 checkpoint signaling) under everolimus is a potential **resistance mechanism**, creating a tumor milieu less susceptible to immune-mediated killing. Combination of everolimus with immune checkpoint inhibitors (anti-PD-1/PD-L1) is being explored to counteract this effect. In summary, everolimus downregulates pro-immunity pathways (T-cell activation) and can upregulate immune evasion signals, which is relevant for TNBC or immune-rich tumors (resistance context).

(The above pathways illustrate that everolimus has multi-faceted effects: it directly suppresses oncogenic signaling and cell proliferation pathways, while inducing adaptive pathways like autophagy, MYC, and immune evasion that may lead to drug resistance.)

Upregulated Pathways (upon Everolimus exposure)

Adaptive upregulation of certain pathways has been observed in response to everolimus, either in treated patients, tumor models, or resistant cell lines. These upregulated pathways can reflect the tumor's compensatory mechanisms to counteract mTORC1 inhibition. Key pathways found to increase (transcriptionally or in activity) with everolimus (by context) include:

- **Autophagy Pathway – Upregulated:** Everolimus reliably induces autophagy due to mTORC1 inhibition (mTORC1 is a master negative regulator of autophagy). For example, breast cancer cells treated with everolimus showed elevated LC3-II and autophagosome formation, indicating robust autophagy activation ²⁶ ²⁸. This upregulation is seen in **drug-exposed models** (ER+ cell lines under long-term everolimus) as a survival mechanism. Clinically, increased autophagy gene signatures in post-treatment tumor biopsies would suggest the tumor is mounting an autophagic response. Upregulated autophagy is context-independent (a general cellular response) but is

particularly noted in **ER+ AI-resistant cells** where it contributes to partial everolimus insensitivity²⁸. (This pathway's upregulation is associated with *resistance*, discussed later.)

- **MYC Signaling – Upregulated:** Acquired resistance to everolimus in ER+ breast cancer has been linked to **upregulation of c-MYC** at both mRNA and protein levels²⁹. In cell models mimicking long-term estrogen deprivation (LTED) and everolimus exposure, MYC oncogene expression and its downstream gene signatures become **enriched**²⁹. This is a compensatory feedback: with mTORC1 suppressed, cells may rely on MYC to drive glycolysis, protein synthesis (via alternate pathways), and cell cycle. Notably, even acute everolimus treatment can trigger a rise in MYC expression in some cells²⁹. **Contexts:** This has been observed in ER+ (*luminal*) breast cancer models (MCF7-derived lines) undergoing chronic mTOR inhibition. Clinically, tumors that become everolimus-resistant often show MYC pathway activation. MYC upregulation is an **escape pathway** (driving resistance when upregulated), detailed under “Resistance pathways – upregulation.”
- **AKT/PI3K Reactivation – Upregulated:** Paradoxically, everolimus can lead to **reactive upregulation of upstream PI3K/AKT signaling** due to loss of mTORC1-mediated feedback inhibition³². In everolimus-treated breast cancer cells and patient tumors, studies have noted increased phosphorylation of AKT (Ser473) and activation of receptor tyrosine kinases such as IGF-1R or HER3 as an adaptive response²⁵. This upregulation is often seen in **non-responders or resistant tumors**: for instance, patients whose tumors did not respond to everolimus had higher levels of p-AKT upon treatment, indicating the pathway was trying to compensate. Contexts include *HR+ breast cancers with PIK3CA wild-type*, where alternative feedback loops strongly activate AKT, and *HER2+ cancers*, where HER2/HER3 signaling can surge when downstream is blocked. **Subtype:** This upregulation is not subtype-limited, but is commonly reported in luminal tumors (which frequently have PI3K feedback loops) and HER2-driven tumors. It represents an attempt by cancer cells to bypass mTORC1 blockade by boosting PI3K/AKT (a *resistance-associated* upregulation, also discussed later).
- **ERK/MAPK Signaling – Upregulated (in some contexts):** Although not as universally observed as AKT feedback, there is evidence that **RAS/MAPK pathway** activity can increase to compensate for mTOR inhibition. For example, in everolimus-resistant cell lines, gene expression profiling sometimes shows upregulation of MAPK pathway genes or activation of alternate RTKs (EGFR, FGFR) that signal through RAS-ERK. This is especially plausible in **TNBC or HER2+** contexts where multiple signaling pathways crosstalk. Clinically, *HER2-positive tumors with co-activation of MAPK* tend to derive less benefit from mTOR inhibitors, suggesting that upregulation of MAPK signaling (via KRAS mutations or growth factor ligand feedback) may occur in non-responders. While direct evidence in patients is limited, preclinical data support that if PI3K/mTOR is blocked, some tumors ramp up MAPK to sustain proliferation. (E.g., everolimus combined with MEK inhibitors showed synergy in models, implying MAPK was upregulated and needed dual block.) **Contexts:** TNBC cell models with activated EGFR have shown partial evasion of everolimus via MAPK; similarly, some HER2+ cell lines increase MAPK signaling when mTOR is inhibited. This pathway upregulation is inferred to contribute to resistance. (*Note: Due to limited direct clinical validation, MAPK upregulation is a suspected but not definitively proven adaptive response in everolimus-treated breast cancer.*)
- **PD-L1 and Immunosuppressive Signals – Upregulated:** In the tumor microenvironment, everolimus can induce changes that upregulate immune inhibitory signals. Preclinical **TNBC models** (4T1 murine TNBC) demonstrated that everolimus treatment led to **increased PD-L1 expression** on

tumor endothelial cells ³⁴ ³⁵. This resulted in reduced T-cell tumor infiltration (an immunosuppressive effect). While this specific finding is from mouse models, it suggests that in human tumors, everolimus might upregulate pathways like PD-L1, IDO, or other immunosuppressive cytokines as an adaptive response. Clinically, patients on everolimus have shown lymphopenia and reduced immunoglobulin levels, reflecting systemic immunosuppression ². In tumor tissue, an increase in **immune checkpoint pathway transcripts** could occur, especially in immune-rich cancers. **Context:** Primarily relevant in *TNBC (basal-like)* where immune surveillance is crucial; an upregulation of PD-L1 or related pathways would confer an immune-evasion advantage (potentially making the tumor less sensitive to therapy unless checkpoint blockade is added). This is categorized as a resistance-promoting upregulated pathway.

- **Angiogenesis Pathway (VEGF) – (Transients):** Interestingly, while everolimus generally downregulates angiogenesis (see below), some short-term compensatory effects could include upregulation of other pro-angiogenic factors. For instance, if VEGF is suppressed, tumors might upregulate FGF2 or ANGPTL factors as a bypass. However, evidence in breast cancer of *increased* angiogenic signaling post-everolimus is sparse – most data show a decrease in VEGF. One nuance: hypoxia resulting from vessel pruning by everolimus could transiently increase HIF-1α *stabilization* in areas if drug doesn't fully suppress it, leading to a localized upregulation of some hypoxia-induced genes (though on balance HIF-1α is inhibited by mTOR blockade ²²). **Context:** Not well documented in breast cancer patients; more of a theoretical consideration that tumors might seek alternative angiogenesis routes.

Overall, *pathways upregulated with everolimus exposure* often represent the tumor's escape circuits – **autophagy, MYC, AKT/MAPK, immune evasion** – especially in **non-responders or resistant subtypes (e.g. TNBC)**. By contrast, **responders tend to show less activation of these feedback pathways**, relying instead on the intended pathway suppression.

Downregulated Pathways

Everolimus causes the **downregulation of numerous pathways** directly or indirectly, as part of its therapeutic effect. Key pathways found to be suppressed (transcriptionally or functionally) in the context of everolimus (with relevant subtype/context notes) include:

- **mTORC1 Signaling & Protein Synthesis – Downregulated:** As intended, everolimus **strongly downregulates mTORC1-driven signaling**. This is evidenced by decreased phosphorylation of mTORC1 substrates (S6K, 4E-BP1) and broad suppression of protein translation ¹⁸. The **HALLMARK_MTORC1_SIGNALING** gene set (which includes many ribosomal and translation machinery genes) is significantly repressed in everolimus-treated samples. **Context:** This occurs in all everolimus-exposed breast cancer cells (pan-breast effect), since it's the primary mechanism. It's seen in **responders**: tumors responding to everolimus often exhibit reduced expression of proliferation/metabolic genes that are downstream of mTORC1. This pathway downregulation correlates with tumor growth arrest.
- **Cell Proliferation/Cell Cycle Pathways – Downregulated:** Everolimus leads to **G₁ cell-cycle arrest**, so pathways such as **E2F targets, G₂/M checkpoint, MYC targets, and DNA replication** are downregulated. In endocrine-resistant ER+ cell models, everolimus treatment reduced *Cyclin D1* and

E2F1 levels, and cells accumulated in G₁ ²¹. Gene sets like **KEGG_CELL_CYCLE** or **GO:0045787 (positive regulation of cell cycle)** are suppressed. *Context:* Across subtypes, any tumor that is sensitive to everolimus will show reduced Ki-67 and cell-cycle gene expression. Clinical biopsies from everolimus responders in BOLERO-2 showed drops in proliferation markers (consistent with this pathway's downregulation). This is a **mechanism-related** pathway suppression (desirable for efficacy).

- **Estrogen Receptor Signaling – Downregulated:** In HR+ cancers, everolimus has been shown to **downregulate ERα signaling**. ERα mRNA and protein were markedly decreased in AI-resistant MCF7-derived cells upon everolimus exposure ²⁰. Consequently, estrogen-responsive gene sets (e.g., **HALLMARK_ESTROGEN_RESPONSE_EARLY/LATE**) are downregulated. *Context:* This is specific to ER+ (luminal) breast cancer. In everolimus + endocrine therapy clinical settings, a decrease in ER-related transcripts (PGR, GREB1, etc.) was observed, reflecting restoration of hormone control. Downregulation of the estrogen pathway is a **therapeutically beneficial effect** – it signifies that the tumor's ER-driven growth is being curtailed. Notably, in some cases everolimus can restore hormone sensitivity by inhibiting pathways that cross-talk to ER, effectively “reining in” ER signaling ²⁷. (The converse is also true: if a tumor loses ER expression entirely, that downregulation of ER makes endocrine therapy irrelevant and can cause resistance to the combination.)
- **DNA Damage Repair (DDR) Pathways – (Potentially Downregulated):** mTOR links to DNA repair by regulating the translation of repair proteins. Everolimus might indirectly reduce expression of some DNA repair genes (e.g., BRCA1, RAD51) by inhibiting protein synthesis. There isn't strong direct evidence in breast cancer patients that DDR gene sets are broadly down, but some **GO DNA_REPAIR** categories could be suppressed as a downstream effect of cell-cycle arrest. *Context:* If present, this would be general and not subtype-specific. This is not a primary known effect, so we note it as a possible minor downregulation in the context of decreased proliferation (cells in G₁ may not actively engage some DNA repair programs).
- **Angiogenesis/VEGF Pathway – Downregulated:** Everolimus exhibits anti-angiogenic effects by **downregulating HIF-1α and VEGF signaling** ^{22 23}. In a clinical study (outside breast cancer, but mechanistically relevant), everolimus reduced microvessel density and VEGF expression in tumors. In breast cancer, one can infer that gene sets like **HALLMARK_ANGIOGENESIS** or **REACTOME_VEGF signaling** would be downmodulated in responders. Indeed, in endocrine-resistant breast tumors which often express angiogenic factors, mTOR inhibition lowers pro-angiogenic cytokines (possibly contributing to tumor stasis). *Context:* This is likely in **HR+ tumors** too, since those can be richly vascularized; it could also apply in **HER2+** (HER2 upregulates VEGF) and **TNBC** (often highly angiogenic). Angiogenesis downregulation is part of the **mechanism of action** that may not be clinically obvious (no specific biomarker tracked), but it aligns with everolimus's known anti-angiogenic properties ²².
- **Immune Activation Pathways – Downregulated:** Everolimus's immunosuppressive nature means pathways related to T-cell proliferation and function (e.g., **IL-2/STAT5 signaling, T-cell receptor signaling**) are downregulated. In patients, everolimus is known to reduce mitogen-driven T-cell expansion ³³. Though tumor biopsies are rarely assayed for this, one might see decreased expression of genes like *Granzyme B*, *IFN-γ*, *CD8A* in the tumor microenvironment under everolimus treatment (if immune cells are present). *Context:* Particularly relevant in immune-infiltrated tumors (e.g., some TNBCs). This downregulation is a double-edged sword: it can reduce antitumor immunity

(a **resistance risk**), but also might reduce inflammation-driven growth signals in the tumor. Regardless, it is an observed effect (from transplant medicine and extrapolated to cancer): **T-cell and cytokine pathways are suppressed** ³³.

- **Growth Factor/HER2 Signaling – Downregulated (Context-specific):** In HER2-amplified tumors, everolimus can indirectly downregulate some HER2-driven output by interrupting downstream PI3K signaling. For instance, genes in the **ERBB2 signaling** cluster might be less expressed as proliferation slows. However, everolimus does *not* directly downregulate HER2 protein levels (and in fact HER signaling can remain active, leading to AKT feedback). Still, in tumors where everolimus has effect, one finds **downstream HER2 pathway genes (like cell-cycle targets)** reduced. *Context:* HER2+ patients in the BOLERO-3 trial had improved PFS with everolimus + trastuzumab, implying effective pathway suppression in at least a subset ³⁶. In those responders, one could infer that HER2-PI3K signaling output was curtailed (though HER2 itself wasn't eliminated). This is a nuanced point: we consider that effectively everolimus *contributes* to downregulating HER2 pathway *activity* (not receptor levels) by cutting off its main signaling arm.

In summary, *pathways downregulated by everolimus* align with its therapeutic intent: **proliferation, ER signaling, mTORC1 activity, and angiogenesis are diminished**, especially in responders ²⁴ ²⁰. These changes are usually **beneficial**, correlating with tumor shrinkage or stabilization. Notably, an absence of expected pathway downregulation (e.g., if a tumor fails to reduce Ki-67 or mTORC1 activity on treatment) might indicate intrinsic resistance.

Sensitivity Pathways for Upregulation

These are pathways whose **upregulation (high baseline activity or induced expression)** is associated with **increased sensitivity** to everolimus in breast cancer. In other words, if a given pathway is more active in a tumor, that tumor is *more likely to respond* to everolimus, often because the drug specifically counteracts that active pathway (creating a dependency). Key “upregulated = sensitive” pathways include:

- **PI3K/AKT/mTOR Pathway (Upregulated)** – Perhaps the clearest determinant of sensitivity is an **upregulated PI3K-AKT-mTOR signaling network**. Tumors with high basal activity of this pathway tend to be *addicted* to it, so inhibiting mTORC1 yields a stronger effect. Clinically, HR+ breast cancers with **PIK3CA mutations or PTEN loss** – lesions that hyperactivate PI3K/AKT – showed improved outcomes on everolimus in some studies ⁹ ⁴. For example, the BOLERO-2 trial subgroup analyses suggested a trend (though not statistically definitive) of greater PFS benefit in patients with PI3K pathway alterations. Similarly, TNBC tumors often have PTEN loss leading to AKT/mTOR activation; in a phase II trial, those TNBCs derived notable benefit from adding everolimus ⁹ ⁷. Thus, an “up” PI3K/AKT pathway (high p-AKT, high mTOR activity signature) correlates with **everolimus sensitivity**. *Context:* pan-breast cancer, but especially **luminal A/B tumors with PIK3CA mutations and PTEN-null TNBC**. These tumors’ reliance on the PI3K-mTOR node makes them vulnerable to mTORC1 blockade.
- **Estrogen Receptor Signaling (Upregulated)** – In HR+ cancers, robust ER signaling (high estrogen-responsive gene expression) can indicate that the tumor, while ER-driven, has engaged the PI3K/mTOR pathway to grow (since pure ER activity was controlled by prior endocrine therapy). Everolimus is particularly effective in **endocrine-resistant, ER-positive tumors** where ER is still expressed (up)

but not sufficient for growth – instead, PI3K/mTOR is co-activated. In such cases, strong ER pathway activity means the tumor remains at least partly hormone-driven, and by inhibiting mTOR and continuing endocrine therapy, one-two punch can be delivered. This was evidenced by everolimus restoring endocrine sensitivity in AI-resistant cell lines that still had upregulated ER signaling ²⁷ . Patients whose tumors retained ER expression (and had high ER-regulated gene profiles despite AI therapy) responded better to everolimus + exemestane in some analyses (the **BOLERO-2 trial** noted benefit largely in ER-dependent tumors that had acquired PI3K pathway activation ⁴). Therefore, paradoxically, **high ER pathway activity plus PI3K activation** is a context of vulnerability – the tumor is reliant on both, and everolimus helps shut down the non-ER escape pathway while endocrine therapy targets ER. *Context: HR+/HER2-, luminal breast cancers after failure of nonsteroidal AI, with persistent ER expression. (If ER pathway is completely lost, this sensitivity is not applicable.)*

- **HER2/EGFR Signaling (Upregulated)** – Tumors with **upregulated HER2 or EGFR signaling** can be sensitive to everolimus *when combined with HER2-directed therapy*. For instance, the BOLERO-3 trial in HER2+ breast cancer found that tumors with activated HER2/IGF1R pathways that had developed resistance to trastuzumab could benefit from everolimus addition ³⁶ . Especially in **HER2-positive, hormone receptor-negative** cases, high HER2 signaling makes the tumor highly PI3K/mTOR-dependent (since HER2 activates that pathway strongly). Everolimus upsets this dependency. Subgroup data from BOLERO-1 showed that the *HER2+/HR- subset* (where HER2 signaling is “up” and unopposed by ER) derived a ~7-month PFS improvement from everolimus ⁶ . This suggests that **upregulated HER2 pathway activity** (e.g., high HER3, high p-S6 in HER2+ tumors) predicts sensitivity to adding mTOR inhibition on top of anti-HER2 therapy. *Context: HER2-enriched subtype (HER2+, ER-).* Note that in HER2+/ER+ (luminal HER2) tumors, the presence of ER confounded the benefit, possibly requiring simultaneous endocrine therapy. But broadly, a tumor with a **high-growth-factor-signaling signature** (HER2, IGF) that funnels into mTOR can be vulnerable to everolimus.
- **“Proliferation/Cell-Cycle High” (Upregulated Ki-67 and Cyclin D)** – Tumors that are highly proliferative (grade 3, high Ki-67) might paradoxically be more sensitive to cell-cycle arrest by everolimus. A fast-growing tumor with upregulated E2F/cell-cycle pathways stands to be impacted when mTORC1 is inhibited (as protein synthesis of cyclins drops). Clinically, this is nuanced: often high Ki-67 indicates aggressive biology that might respond to chemo; however, retrospective analyses (e.g., 4EVER and BOLERO-2 extension studies) suggested that patients with **high Ki-67** had a substantial PFS gain with everolimus, possibly because those tumors had active mTOR driving proliferation ³⁷ . *Context: across subtypes, but especially luminal B (ER+ but high proliferation) cancers.* In these, the combination of endocrine + everolimus was beneficial, indicating that upregulated proliferation pathways made them dependent on mTOR signals which everolimus cut off. So, a “high proliferation signature” could be a marker of sensitivity to mTOR inhibition.
- **Immune-Inflamed Microenvironment (Upregulated Immune)** – This is speculative, but an **immune-hot tumor** (with upregulated interferon-γ, cytotoxic genes) might initially respond well if everolimus causes tumor cell stasis, because the immune system can contribute to tumor control. For example, some TNBCs with high TILs (tumor-infiltrating lymphocytes) might be sensitive in the short term. However, since everolimus also dampens immune cells, this is a double-edged sword (discussed under resistance). There isn’t strong evidence that “more immune up = more sensitive” for everolimus; if anything, immune-rich TNBC might benefit more from immunotherapy. Thus, we will not count this firmly as a sensitivity factor. It’s more relevant to combination strategies (e.g.,

upregulated immune pathways necessitate PD-1 inhibitor addition to maximize everolimus efficacy as seen in preclinical combos ³⁵).

In summary, **pathways that when upregulated make tumors highly dependent on mTOR signaling tend to confer sensitivity to everolimus**. High PI3K/mTOR activity (via mutations or receptor signaling) is the clearest such pathway ⁹ ⁴ . This often coexists with hormone signaling in HR+ cancers or HER2 signaling in HER2+ cancers, meaning those contexts (PI3K-up + either ER or HER2 driven) are where everolimus produces the greatest benefit.

Sensitivity Pathways for Downregulation

These are pathways whose **downregulation or loss of function** is associated with increased sensitivity to everolimus. In other words, if a pathway is suppressed in the tumor, the tumor becomes more reliant on the remaining mTOR pathway or otherwise becomes easier to treat with mTOR inhibition. Evidence for downregulation conferring sensitivity is less abundant than for upregulation (since most predictive biomarkers are activations), but a few considerations:

- **Loss of Redundant Pathways (Downregulation of bypass signals)** – Tumors that lack alternative survival pathways tend to be exquisitely sensitive to mTOR blockade. For example, a breast cancer with **low MAPK activity** (no RAS/RAF activation) is essentially fully dependent on PI3K/AKT for proliferation. In such a scenario, everolimus has an open field: there is no strong bypass via MAPK. While not explicitly tested in trials, one can infer that if a tumor has **downregulation of RAS/MAPK signaling**, it might respond better to mTOR inhibition. Conversely, if MAPK is up (as a bypass), as noted, resistance occurs. Thus, *“absence of MAPK upregulation”* (i.e., effectively downregulated relative to other tumors) can be a marker of sensitivity. *Context:* Possibly in **luminal cancers** that don’t have RTK mutations – they rely mainly on PI3K/ER and thus are very sensitive to PI3K or mTOR inhibitors. (This is indirectly supported by data that PIK3CA-mutant/AKT1-mutant ER+ cancers, which usually do not have RAS pathway aberrations, responded to PI3K/mTOR targeting.)
- **PTEN Loss (Downregulation of PTEN)** – This might sound counterintuitive because PTEN loss *activates* PI3K (so it could be seen as an “up” in PI3K pathway, already covered). But from another angle: PTEN is a negative regulator of PI3K; **downregulation or loss of PTEN** means the tumor’s only brake on PI3K is gone, making it extremely mTOR pathway-dependent *and* potentially more sensitive to mTORC1 inhibition (since the tumor cells are “addicted” to that hyperactive pathway). Some preclinical data in breast cancer support that PTEN-null cells are vulnerable to mTOR inhibitors (though they can also activate AKT strongly, requiring adequate drug dosing). Clinically, in the TNBC carboplatin+everolimus trial, many tumors were PTEN-null and benefitted from mTOR inhibition ⁹ ⁷ . So **downregulation of PTEN (a tumor suppressor pathway)** can be viewed as a “sensitivity via dependency” situation. *Context:* **TNBC** and some **HER2+** (HER2 and PTEN loss co-occur in a subset) – those double-pathway aberrant tumors often respond to mTOR blockade plus other therapy, albeit the evidence is still emerging.
- **Lack of mTORC1 feedback loops (Downregulation of S6K feedback)** – If a tumor inherently has low S6K/IRS1 feedback (for instance, due to low IRS1 expression), it might not mount the AKT rebound when mTORC1 is inhibited. This could make it more sensitive to everolimus because one major resistance mechanism (AKT reactivation) is muted. This is a highly specific scenario: some

tumors have downregulated insulin receptor substrate (IRS1) or other feedback components. There isn't direct patient data for this, but conceptually a tumor with **impaired feedback signaling** remains in a suppressed state under mTOR inhibition. *Context:* Possibly in certain **ER+ cancers** where prior long-term endocrine therapy or PI3K inhibition altered feedback loops. Without robust feedback, everolimus effect is more durable – thus downregulation of feedback regulators is beneficial. (This is somewhat theoretical; not used clinically as a biomarker.)

- **BRCA1/2 or HR pathway Deficiency (Downregulation of DNA repair)** – If a tumor is HR-deficient (e.g., BRCA1-mutated TNBC), it might rely on alternate survival pathways which could include PI3K/mTOR for cell cycle checkpoint adaptation. Some research suggests that PI3K inhibitors synergize with PARP inhibitors in BRCA-mutant cancers. By analogy, a *downregulated homologous recombination repair* pathway might make the tumor cells more susceptible to any cell cycle-targeting agent like mTOR inhibitor. However, everolimus is not known to specifically exploit HRD like PARP inhibitors do. So while one might hypothesize HRD tumors (BRCA1/2 mutant) could be slightly more sensitive to the growth-arrest effects of everolimus (since they can't easily repair damage and might undergo apoptosis upon growth arrest), this is **not strongly evidenced in clinical data**. For instance, BRCA status was not a criterion in everolimus studies, and no clear differential effect has been reported. Thus, we consider this largely *insufficient evidence*.

In practical terms, “**sensitive when pathway X is low**” is **less of a paradigm** for everolimus; it's more about what's high. The notable exception is that **absence of alternative drivers** (i.e., a tumor that is a uni-dimensional PI3K/mTOR-driven tumor) is very sensitive. So you could say a *low level of parallel pathways = sensitivity*. For example, a tumor that is **PI3K-driven but “RAS/MEK-low” and “immune-cold”** (so no immune or MAPK interference) could respond extraordinarily well to everolimus because all its eggs are in the mTOR basket.

In summary, while direct “sensitivity via downregulation” pathways are few, one guiding principle is: **the fewer backup pathways a tumor has (the more “single-pathway-addicted” it is), the more an mTOR inhibitor will cripple it**. This often correlates with *downregulation* (or lack) of other pathways such as MAPK.

(If evidence is insufficient for specific downregulated sensitivity pathways, one may conclude that no definitive biomarkers in this category are established – and indeed, in clinical practice, we identify sensitive tumors more by presence of PI3K activation than by absence of something else.)

Resistance Pathways for Upregulation

These are pathways whose **upregulation** is associated with **resistance or reduced response** to everolimus. In breast cancer, multiple compensatory or escape pathways can be activated, leading to therapeutic failure. Proven resistance-associated upregulated pathways include:

- **MYC Oncogenic Pathway – Upregulation Confers Resistance:** As discussed, everolimus-resistant ER+ breast cancers often show **marked upregulation of MYC**. Both acute and long-term everolimus exposure can drive MYC mRNA/protein increases ²⁹. Functionally, high MYC levels allow cancer cells to maintain metabolic and cell-cycle progression even with mTORC1 inhibited (MYC can drive transcription of ribosomal RNAs, tRNAs, and glycolytic enzymes independent of mTOR). *Evidence:* In a study by Bihani et al., depletion of MYC resensitized resistant cells to everolimus, proving MYC

upregulation was causative in resistance ³⁰. Clinically, tumors that did not benefit from everolimus had higher MYC gene signatures ²⁹. Upregulation of MYC can occur via epigenetic changes – everolimus-resistant cells had increased BRD4 binding to the MYC gene, enhancing its transcription ³¹. **Context:** This is primarily observed in **ER+/HER2- (luminal)** breast cancers under mTOR inhibitor pressure. It suggests that if a tumor adapts by turning the MYC pathway “on,” it will likely be resistant. MYC upregulation is thus a key **resistance pathway**; its presence (e.g., high MYC IHC or gene expression) after treatment is a red flag for therapeutic failure.

- **AKT/PKB and Upstream RTKs – Upregulation Drives Resistance:** One of the classic resistance mechanisms to rapalogs is **reactivation of AKT via upregulation of upstream signaling**. Everolimus causes loss of feedback inhibition, so tumors often respond by **upregulating insulin-like growth factor 1 receptor (IGF1R)**, other receptor tyrosine kinases, or by amplifying ligand production (IGF-1, heregulin, etc.), leading to AKT rephosphorylation ³². This “AKT hyperactivation” has been documented in everolimus resistance: e.g., Carew et al. (2011) found that breast cancer cells chronically exposed to RAD001 had **increased AKT phosphorylation** (and also activated alternate PKC family members) ²⁵. Clinically, **tumors with PIK3CA wild-type** (which may rely more on RTK signals) sometimes had less benefit, possibly due to such feedback activation. Additionally, **HER2 signaling upregulation** can also be considered here – e.g., a HER2+ tumor might upregulate HER3 (a dimerization partner) when mTOR is inhibited, reactivating PI3K and AKT. All these represent an “upregulation of proximal signaling” leading to maintained downstream signals despite everolimus. **Context:** **Any subtype**, but especially **HER2+** (HER-family upregulation) and **luminal with growth factor co-drive**. For instance, in BOLERO-1, HR+HER2+ patients (who didn’t gain PFS from everolimus) often had tumors that likely used alternate pathways – possibly indicating that simply blocking mTORC1 led to compensation via the HER pathway (which was not fully shut down without endocrine therapy). Summarily, **if AKT or RTK activity goes up** on treatment, the tumor is effectively resistant.

- **RAS/MAPK Pathway – Upregulation Bypasses mTOR:** Activation of the RAS–RAF–MEK–ERK pathway can allow cells to proliferate independently of mTOR signals. There is evidence in other cancers that rapalog resistance involves MAPK upregulation; in breast cancer, this has been suggested by preclinical models. For example, everolimus-resistant MCF7 cells were found to have elevated ERK phosphorylation, indicating a switch to MAPK drive. Upregulation of **EGFR or FGFR** (which signal through RAS/MAPK) in response to mTOR inhibition has been observed in some lab studies. If a breast tumor has a **KRAS or BRAF mutation** (rare in breast, but ~5% of TNBC might have RAS pathway mutations), it likely will not respond well to everolimus, as the MAPK pathway is constitutively up and can carry the load of driving proliferation. Similarly, **upregulation of MEK/ERK** via increased growth factor ligands (like heregulin activating HER3→RAS) could cause resistance. **Clinical hint:** In one case series of everolimus in metastatic ER+ cancer, tumors that progressed quickly had mutations in MAPK pathway genes, suggesting that up-front MAPK activation made mTOR inhibition ineffective (since MAPK was an alternate driver). Thus, we include **MAPK pathway upregulation** as a likely resistance pathway when upregulated. **Context:** Possibly more relevant in **HER2+** (since HER2 also activates RAS/MAPK) and **TNBC** (where EGFR can be overexpressed). It’s an area of active research (e.g., combining everolimus with MEK inhibitors is being explored to counter this escape).

- **Survival Pathways – NF-κB/Anti-apoptotic Signals Up:** Some studies have noted that everolimus-resistant cells overexpress anti-apoptotic proteins (e.g. BCL-2, Survivin (BIRC5)). For instance, one

report indicated **survivin was upregulated** in everolimus-resistant breast cancer cells, contributing to resistance (cells avoid death) ³⁸ . Upregulation of the **NF-κB pathway** can lead to increased transcription of such survival genes. If mTOR is inhibited, some cells may compensate by activating NF-κB (since mTORC1 normally can suppress some aspects of NF-κB activation, its inhibition in immune cells actually can enhance NF-κB, but in cancer context it's not well-characterized). In any case, a tumor that **upregulates anti-apoptotic and stress response pathways** can resist the cytostatic stress of everolimus. *Context:* Could be seen in **ER+ cells** with chronic exposure (as a general stress survival response) or **TNBC cells** engaging inflammatory loops. While not as widely reported as MYC or AKT, it's logically consistent that if a tumor increases expression of survival factors (BCL2, MCL1, etc.), it will better withstand the growth inhibitory signals – thus representing a resistance mechanism.

- **Immune Evasion Signals – Upregulation:** As noted earlier, everolimus can induce immunosuppressive changes – an increase in PD-L1, cytokines like IL-10, etc. **Upregulation of PD-L1** on tumor or stromal cells would help the tumor hide from any T-cell attack, potentially making the cancer persist despite therapy (since everolimus alone mostly causes tumor stasis, the immune system is needed to clear cells). In the TNBC 4T1 study, **PD-L1 upregulation under everolimus** directly led to resistance that was overcome by anti-PD-1 therapy ³⁴ ³⁵ . Translating to humans: if a breast tumor under treatment shows increased PD-L1 expression (e.g., on repeat biopsy), it might indicate an evolving immune-evasive resistance. Similarly, **upregulation of pro-angiogenic factors** could be considered – if a tumor ramps up VEGF or finds new vessel growth pathways (though everolimus tends to reduce VEGF initially, a highly hypoxic tumor might later secrete more of other factors).

In essence, **resistance to everolimus is often characterized by “up” signals that bypass the need for mTOR** – whether it's another oncogene like MYC, a parallel pathway like MAPK or AKT, or supportive pathways like NF-κB and immune evasion. Monitoring for these upregulated changes (e.g., rising p-AKT or MYC levels) could potentially identify resistance early. Combination therapies targeting these upregulated pathways (e.g., adding AKT inhibitors, CDK inhibitors for Myc, or checkpoint inhibitors for PD-L1) are logical strategies suggested by these findings ³¹ ³⁵ .

Resistance Pathways for Downregulation

These are pathways whose **downregulation or loss** leads to resistance to everolimus. This can include loss of the very targets or co-factors that everolimus relies on for effect, or loss of tumor dependencies that renders the drug's action less relevant. Notable (though fewer) examples:

- **Estrogen Receptor Pathway – Downregulation Causes Resistance:** In HR+ breast cancer, if the tumor *downregulates or loses ER expression* (becoming functionally ER-negative), it likely indicates a shift to a different phenotype that may not benefit from everolimus plus endocrine therapy. Clinical evidence: In the BOLERO-1 trial for HER2+ disease, the subset that benefited from everolimus was the ER-negative group ⁶ . This suggests that in ER-positive/HER2+ tumors, once they **downregulated ER signaling (or effectively bypassed it)**, adding everolimus without endocrine therapy had little impact. More directly, in HR+ metastatic breast cancer, some patients' tumors lose hormone receptor expression after successive therapies – such tumors are less endocrine-driven and possibly rely on other pathways (like growth factors or MAPK). Everolimus's approved use is with

endocrine therapy; if the target pathway (ER) is gone, the combination is far less effective. Therefore, **loss or downregulation of ER (and related hormone pathways)** constitutes a mechanism of resistance. It means the tumor has likely shifted to a state where mTOR is being driven by alternate signals or the tumor is too aggressive (perhaps neuroendocrine differentiation, etc.). *Context: Luminal tumors transitioning to hormone-independent status.* A practical example is if an initially ER+ tumor comes back ER-negative on a biopsy after AI therapy – using everolimus + AI in that scenario would be futile; the tumor's ER pathway is down (and that downregulation confers resistance to the prior hormonal strategy and by extension the combination strategy).

- **mTOR Pathway Itself Downregulated (Loss of Target) – Resistance:** While rare, a tumor could acquire mutations or alterations that make it less dependent on mTORC1. For instance, if a tumor cell **downregulates mTOR expression** or key components of that pathway (S6K, 4EBP1), it might ironically become less sensitive to everolimus because it no longer heavily requires mTORC1 for survival – it may have switched to alternate routes like MAPK or autophagy entirely. There's no known clinical scenario where mTOR is lost (since that would generally be detrimental to the cell), but there are reports in other cancers of mutations in mTOR that reduce drug binding (which is not downregulation per se, but an alteration of the target). In the context of “downregulation”, one could consider **loss-of-function in TSC1/TSC2** (which normally restrain mTOR) – that *activates* mTOR (so not a downregulation of pathway; it's opposite). So strictly speaking, tumors don't resist by downregulating the target pathway – they resist by **upregulating alternate pathways** (covered above) or by **altering the target** (mutations in MTOR or in FKBP12-binding domain, observed in some everolimus-resistant cancers like renal cell carcinoma, though not yet reported in breast). If such a mutation happened, effectively the **functional mTOR signaling might be reduced in how it interacts with the drug**, making the drug ineffective (the pathway appears “down” to the drug's effect).
- **Loss of PTEN (counterintuitive)** – Here we revisit PTEN: PTEN loss makes tumor initially sensitive (due to dependency as discussed), but there is data from other cancers (e.g., metastatic renal cell carcinoma) that **extreme activation of PI3K (via PTEN loss) can cause resistance to rapalogs** because the mTORC1 blockade is not enough – the tumor may activate mTORC2 or some mTOR-independent AKT outputs. In breast cancer, however, PTEN loss generally correlates with at least some sensitivity to everolimus (as seen in TNBC trial) ⁹. So PTEN “downregulation” we counted as sensitivity. There isn't a clear example of a pathway that, when suppressed, causes resistance aside from losing the entire dependency on the targeted mechanism (like losing ER as above, or losing reliance on mTOR by switching to MAPK, which again is an upregulation of MAPK, not a downregulation).

Given these considerations, we conclude that **few pathways, when downregulated, directly cause resistance** – rather, resistance is about finding other pathways. If forced to name one, **loss of ER signaling** is a prime example in the HR+ context as described. Another possible one: **downregulation of immune surveillance** (i.e., a tumor becoming “immune-cold”) could make everolimus (or any therapy) less effective if part of its effect relied on immune-mediated tumor cell clearance. Everolimus itself can contribute to that by reducing T-cell infiltration, essentially the tumor microenvironment becomes downregulated in immune activity, and that state is associated with poorer outcomes. In immunotherapy studies, loss of TILs is bad for checkpoint inhibitors; analogously, if a tumor on everolimus loses its TILs (immune down), it might progress. However, this is a secondary effect and not well documented in patients.

In summary, “**resistance via downregulation**” is not a prominent theme for everolimus in breast cancer. Instead, resistance mostly comes from turning *other* pathways up, not turning things off. The main scenario is the tumor turning off the hormone receptor pathway (in HR+ disease), which leaves everolimus without its synergistic partner (endocrine therapy), leading to treatment failure. If evidence or clear examples are lacking, one can state: *Evidence of pathways whose loss confers resistance to everolimus is insufficient; resistance is predominantly driven by activation of alternate pathways rather than loss of specific pathways.* ³⁹ ⁴⁰

(*Insufficient/Conflicting Evidence Note:* Indeed, aside from HR status changes, no specific “downregulated pathway biomarker” is used clinically to predict everolimus resistance. Everolimus resistance is better defined by what goes up (AKT, MYC, etc.) than what goes down.)

Breast Cancer Subtype–Stratified Evidence

Everolimus’s effects and clinical utility vary across **breast cancer subtypes**, each with distinct molecular drivers:

- **HR+/HER2– (Luminal A/B) Subtype:** This is the subtype where everolimus has an established role. Postmenopausal women with HR+ HER2– advanced breast cancer that progressed on nonsteroidal AIs benefited from everolimus + exemestane, as shown in the BOLERO-2 trial ⁴¹ ³. In luminal tumors, endocrine resistance often involves PI3K/mTOR activation ⁴. Everolimus combined with endocrine therapy targets that mechanism – significantly prolonging PFS (from ~3–4 months to ~7–11 months median) ⁴². **Biomarker evidence:** While initial hopes were that PIK3CA mutations would predict benefit, in BOLERO-2 even PIK3CA–wildtype patients benefited, so no mandatory biomarker is required. However, retrospective data indicated numerically longer PFS in PIK3CA-mutant tumors (not significant), and possibly greater benefit in tumors with low PR (a sign of PI3K pathway activation) – these are hypothesis-generating. **Endocrine context:** Everolimus is typically used after AI failure; now with CDK4/6 inhibitors as first-line standard, everolimus is often considered after progression on CDK4/6 + AI. There is evidence that everolimus still has activity post-CDK4/6: clinical series have reported meaningful PFS with everolimus after palbociclib, implying non-cross resistance. In HR+ patients without PIK3CA mutation (who are not candidates for alpelisib), everolimus + endocrine (exemestane or fulvestrant) is recommended by guidelines as a subsequent line ⁴³. **Subtype nuances:** Luminal B (higher proliferation) tumors may derive slightly more absolute benefit than Luminal A (indolent) from mTOR inhibition, since the former often harbor PI3K activation and are endocrine-resistant. Overall, everolimus is **most effective in ER+/HER2– disease**, improving outcomes in endocrine-resistant settings and now integrated into NCCN, ESMO, and ASCO guidelines for this subtype ⁴³.
- **HER2-Positive Subtype:** Everolimus is not standard first-line for HER2+ breast cancer, but has been studied in combination with anti-HER2 therapy for resistant cases. The **BOLERO-3 trial** (HER2+, trastuzumab-resistant metastatic breast cancer) demonstrated that adding everolimus to trastuzumab + vinorelbine modestly but significantly improved PFS in patients who had progressed on prior trastuzumab (median PFS 7.0 vs 5.78 months) ³⁶. Particularly, BOLERO-3 and a biomarker analysis suggested the benefit was concentrated in patients with **HR-negative/HER2+ tumors** ⁴⁴. Meanwhile, the **BOLERO-1 trial** evaluated everolimus with first-line paclitaxel/trastuzumab in HER2+ MBC: it was *overall negative*, but the pre-specified HR-negative subset saw an intriguing PFS prolongation (~20.3 vs 13.1 months, HR 0.66) that narrowly missed the significance threshold ⁶.

This indicates that in **HER2-enriched (ER-) tumors**, mTOR inhibition adds value by blocking HER2-downstream PI3K signals (especially since those tumors often have PTEN loss or PIK3CA mutations). In **HER2+/ER+ (Luminal HER2)** tumors, everolimus did not show a clear benefit without concurrent endocrine therapy, likely because the driving resistance pathway was hormonal or because mTORi toxicity limited combination dosing with chemo in that subset. Currently, everolimus is not widely used in HER2+ breast cancer in practice, but **guidelines recognize it as an option in trastuzumab-resistant cases**. For example, NCCN mentions considering everolimus + trastuzumab (± chemo) for heavily pretreated HER2+ patients, referencing BOLERO-3. Additionally, translational studies show that **HER2+ tumors with PI3K pathway aberrations** (like PIK3CA mutations) have inferior outcomes on HER2-targeted therapy; combining mTOR inhibition can counteract that pathway and resensitize tumors to anti-HER2 agents ³⁶. In summary, **HER2+ evidence**: everolimus can overcome resistance mediated by PI3K/mTOR in HER2+ disease, especially when hormonal signaling is not a factor (ER-negative). It's an option for dual-targeted therapy in refractory HER2+ breast cancer, though toxicity (overlapping stomatitis, etc.) must be managed carefully ⁴⁵.

- **Triple-Negative (TNBC) Subtype**: TNBC generally does not have approved targeted therapies that overlap with everolimus's mechanism (except indirectly via PI3K pathway in subsets). Everolimus is not an established treatment in TNBC outside clinical trials. However, TNBC often exhibits **PI3K/AKT pathway dysregulation** (PTEN loss in ~35% of cases, PIK3CA mutations in ~10-15%, AKT1 mutations in a smaller fraction) ⁹. This provided rationale to test mTOR inhibition. A notable trial is the **PreCOG 0105** phase II (2015) which combined everolimus with carboplatin in metastatic TNBC: it showed a clinical benefit, prompting a randomized phase II (NCT02531932). The **recent randomized trial (Patel et al. 2023)** reported that everolimus + carboplatin significantly improved PFS over carboplatin alone (4.7 vs 4.2 months; HR 0.49, $P = 0.039$) ⁸. While the absolute gain was small, the 52% hazard reduction is compelling and suggests a subset of TNBC derives major benefit ⁷. Interestingly, there was no OS improvement, indicating some tumors still behaved aggressively ⁴⁶. The TNBC tumors most likely to benefit were those with **mTOR pathway activation** – e.g., cases with PTEN loss (a common event in TNBC that leads to unchecked AKT/mTOR activity) ⁹. Additionally, preclinical data in TNBC models (including basal-like 4T1 mouse tumors) show everolimus can inhibit tumor growth when combined with immune checkpoint blockade ⁴⁷ ³⁵, hinting at future combination strategies. Currently, everolimus is not a part of standard TNBC regimens (unlike PARP inhibitors for BRCA-mutant TNBC or immune checkpoint inhibitors for PD-L1+ TNBC). However, these findings **suggest a role in subsets**: possibly **PI3K-pathway-addicted TNBC**, or as **chemo-sensitization** (everolimus seems to potentiate platinum efficacy, perhaps by blocking DNA damage response via mTOR – an area being investigated). As of 2025, ongoing trials are likely evaluating mTOR inhibitors in TNBC, potentially with patient selection by PTEN or PI3K biomarkers. So, TNBC evidence is emerging: everolimus can slow TNBC growth in some patients, but the data is early and not yet practice-changing.

- **BRCA-mutated/HRD Subtype**: BRCA1/2-mutant breast cancers (which are often triple-negative) have not been a primary target for everolimus therapy, given PARP inhibitors directly address that biology. That said, there is crosstalk: PI3K/mTOR pathway is frequently activated in BRCA-mutated tumors as well (BRCA1 loss can lead to compensatory PI3K activation). Preclinical synergy has been noted between mTOR inhibitors and PARP inhibitors, suggesting mTOR inhibition might exploit metabolic vulnerabilities in HR-deficient cells. Clinically, there isn't a dedicated trial of everolimus in BRCA-mutant patients, but subset analysis from TNBC studies did not show obvious differences by BRCA status (numbers were small). For HRD-high tumors (beyond BRCA), again no direct data – these

tumors may be sensitive to everolimus if they also have PI3K pathway aberrations, but HRD itself doesn't guarantee PI3K activation. In practice, BRCA-mutated patients would more likely get PARP inhibitors than everolimus. If they exhaust those, and if their tumor shows PI3K/mTOR activation, one might consider an mTOR inhibitor in a trial or off-label context. Overall, **no specific evidence** that BRCA/HRD status predicts a better or worse response to everolimus – it's more about the PI3K pathway status. Therefore, **BRCA/HRD** is not a defined subgroup for everolimus, beyond overlapping with TNBC as discussed.

- **Basal-like vs. Luminal vs. HER2-enriched vs. Claudin-low (intrinsic subtypes):** These finer stratifications echo what's said above. *Basal-like* overlaps with TNBC; they might have PTEN loss and some may respond as per TNBC trials. *HER2-enriched* overlaps with HER2+/HR-, where we saw a signal of benefit in BOLERO-1 ⁶. *Luminal A/B* overlaps with HR+; luminal B (higher proliferation) often has PI3K mutations and benefitted from everolimus in trials; luminal A (lower grade) can still benefit if it develops AI resistance (though some luminal A with low proliferation might not be ideal candidates due to lower absolute benefit vs toxicity). *Claudin-low* tumors are often TNBC with high immune infiltrate and EMT features – they might be less likely to respond to everolimus alone since they have multiple alternate pathways (but could benefit in combinations e.g., with immunotherapy).

In conclusion, **everolimus is most firmly indicated in the HR+/HER2- subtype**, shows **conditional benefit in HER2+ (ER-) disease** and is an **experimental option in TNBC** with PI3K pathway activation. Subtype stratification is important: we always consider ER/HER2 status when evaluating everolimus. For instance, if an ER+ tumor loses ER (switches subtype to basal-like), we expect poor results with everolimus. If a HER2+ tumor is also ER-, adding everolimus might help overcome HER2 therapy resistance. These nuances align with clinical observations: **Hormone receptor expression modulates everolimus efficacy in both HR+ and HER2+ settings** ⁴⁴, and **PI3K pathway activation modulates efficacy in all settings** ⁴. Guidelines reflect these: NCCN lists everolimus for HR+/HER2- after specific lines, and notes its evaluation in HER2+; it's absent in TNBC sections except in trial contexts.

Contraindications and Safety

Everolimus therapy comes with specific contraindications and safety considerations, informed by regulatory labels (FDA/EMA) and clinical experience:

Contraindications:

- **Hypersensitivity:** Everolimus is contraindicated in patients with severe hypersensitivity to everolimus, to other rapamycin derivatives (e.g., sirolimus), or any of its excipients ⁴⁸. Cases of hypersensitivity reactions (angioedema, anaphylaxis, etc.) have been observed ⁴⁸, so any history of hypersensitivity to these agents precludes use.
- **Pregnancy:** Everolimus can cause fetal harm. It is classified as FDA Pregnancy Category D ⁴⁹. Women of childbearing potential must use effective contraception during treatment and for ~8 weeks after the last dose ⁵⁰. If a patient becomes pregnant on everolimus, she should be informed of the potential risk to the fetus ⁵¹ ⁵². In practice, everolimus is *contraindicated in pregnancy* due to its mechanism (inhibiting a key developmental pathway, mTOR, and its immunosuppressive effects).
- **Breastfeeding:** Although not always listed as a “contraindication,” everolimus is **not recommended during breastfeeding**. It is likely excreted in milk (other rapalogs are), and because of potential serious adverse effects in the nursing infant, women are advised to discontinue nursing or the drug. (The FDA label

recommends against breastfeeding for 2 weeks after final dose, given everolimus's half-life and potential for transfer.)

- **Severe Hepatic Impairment:** Everolimus is primarily metabolized by the liver. In patients with severe hepatic impairment (Child-Pugh class C), everolimus exposure is greatly increased. Use in such patients is *not recommended* unless the benefit outweighs risk; if used, dose must be substantially reduced (e.g., do not exceed 2.5 mg daily) ⁵³ ⁵⁴ . Moderate impairment (Child-Pugh B) also requires dose reduction. Careful monitoring of everolimus blood levels and toxicities is mandated in any patient with hepatic dysfunction.

- **Strong CYP3A4 Interactions:** Coadministration of everolimus with strong CYP3A4 inhibitors (e.g., ketoconazole, ritonavir) or inducers (e.g., rifampin, phenytoin) is contraindicated or should be avoided due to significant changes in everolimus levels ⁵⁵ ⁵⁶ . Strong inhibitors can raise everolimus AUC several-fold, risking toxicity; strong inducers can virtually nullify everolimus exposure, risking therapeutic failure. If absolutely necessary, dose adjustments and therapeutic drug monitoring are required ⁵⁵ . (In practice, one would avoid combining with things like voriconazole or rifampin – find alternatives.)

- **Live Vaccines:** Patients on everolimus should not receive live vaccines and should avoid close contact with individuals who have received live vaccines ⁵⁷ ⁵⁸ . Everolimus-induced immunosuppression can increase the risk of disseminated infection from live vaccines. This is a precaution in guidelines: e.g., no intranasal flu, no oral polio, no yellow fever vaccines during therapy.

Safety and Warnings:

- **Non-infectious Pneumonitis:** Everolimus has a known risk of causing non-infectious interstitial pneumonitis. Clinical trials reported ~12-14% any-grade pneumonitis in oncology patients, with ~3-4% grade ≥ 3 ⁵⁹ . Patients may present with dry cough, dyspnea, or hypoxia. Some fatal cases have occurred ⁴⁵ . Monitoring for respiratory symptoms is essential; everolimus should be held or discontinued for grade 2 or higher pneumonitis, and corticosteroids may be required for symptomatic pneumonitis. Baseline lung disease is a caution (everolimus isn't absolutely contraindicated, but those patients need close monitoring).

- **Stomatitis (Mucositis):** Aphthous-like mouth ulcers are very common. In BOLERO-2, ~44% of patients got stomatitis (8% grade 3) ⁵⁹ . This can occur early (within first 2 months). Management includes non-alcoholic mouthwash (dex rinse), topical steroids, and dose interruptions for painful ulcerations. Stomatitis is not a contraindication but frequently necessitates dose adjustment; it's a **dose-limiting toxicity** in practice.

- **Metabolic Effects:** Everolimus can induce **hyperglycemia, hyperlipidemia**. Grade 3 hyperglycemia occurred in ~4-15% of patients in trials ⁵⁹ . It reduces insulin sensitivity and can precipitate or worsen diabetes. Regular monitoring of blood glucose and lipids is recommended. If a patient has uncontrolled diabetes, that should be managed aggressively (it's not an absolute contraindication, but significant hyperglycemia might warrant holding therapy until better controlled).

- **Myelosuppression:** Everolimus can cause anemia (~6% grade 3 in BOLERO-2) ⁵⁹ , and less commonly neutropenia or thrombocytopenia. It's generally mild, but periodic blood counts are advised. Severe hematologic toxicity is uncommon in breast cancer dosing (10 mg daily). If a patient has baseline severe cytopenia (e.g., from bone marrow involvement), use caution.

- **Infections:** As an immunosuppressant, everolimus heightens infection risk. Reports include opportunistic infections (e.g., pneumocystis jirovecii pneumonia, CMV) particularly in transplant doses or oncology patients on multiple therapies. In BOLERO-2, infection rates (mostly mild) were higher on everolimus. Rarely, fatal infections can occur. Patients should be counseled to report fever or other infection signs promptly. Prophylaxis for pneumocystis is not routine in oncology use but can be considered if combined immunosuppression exists.

- **Renal Function:** Everolimus can cause proteinuria and has occasionally led to renal failure in combination with other nephrotoxic drugs. In cancer patients this is not a major issue, but trace proteinuria is observed. If a patient has significant renal impairment, dose adjust (though hepatic metabolism is more relevant).

- **Wound Healing:** mTOR inhibitors can impair wound healing (mTOR is crucial in cell proliferation). Everolimus should be used cautiously around major surgery – usually held a week before and after surgery if possible. NCCN guidelines mention this for surgical contexts.
- **Concurrent Radiation:** There have been reports of increased radiation-induced toxicity (“radiation recall”) with mTOR inhibitors – e.g., an increase in pneumonitis if chest radiation and everolimus overlap. Not a formal contraindication, but a caution.

Population Considerations:

- **Geriatric Use:** No specific contraindication. In BOLERO-2, older patients had similar efficacy. Toxicity (stomatitis, pneumonitis) might be slightly more frequent in ≥ 65 . Start at standard dose but monitor closely.
- **Pediatric Use:** Not relevant for breast cancer (everolimus is used pediatrically in tuberous sclerosis at different doses). For completeness: pediatric patients on everolimus (for SEGA or TSC) have higher risk of infections; growth should be monitored. But in breast cancer, patients are adults.
- **Combination Therapy:** Everolimus is often combined with endocrine therapy. One must consider safety of the combination: e.g., with exemestane, overlapping stomatitis and fatigue were observed but manageable. Everolimus + tamoxifen is not commonly done (potentially higher clot risk from tamoxifen plus mTOR effect on lipids – not studied much). Everolimus + chemo (as in BOLERO-1 with paclitaxel) had increased toxicity: severe neutropenia and stomatitis required proactive management ⁶⁰ ⁶¹. So combinations can be toxic; proactive supportive care (growth factors, dose mods) are needed.
- **Vaccinations:** As noted, avoid live vaccines ⁶². Inactivated vaccines may be given but immune response could be suboptimal. Annual flu shot (inactivated) is recommended.

Monitoring: Due to these safety issues, patients on everolimus require regular monitoring: CBC, metabolic panel (glucose, lipids), liver enzymes, and clinical assessment of mouth and lung symptoms at each visit. Blood drug levels are not routinely monitored in oncology (unlike transplant), but could be considered if unusual toxicity or adherence issues. Therapeutic range from transplant literature is 5-15 ng/mL; oncology often tolerates levels ~20 ng/mL for efficacy, but toxicity rises accordingly.

In summary, **everolimus is a drug with a unique toxicity profile:** mucositis and metabolic effects are signature, and non-infectious pneumonitis is the most serious risk. **Contraindications** are mainly centered on safety concerns (hypersensitivity, pregnancy) and patient selection (severe organ dysfunction, unmanageable comorbidities). With careful patient monitoring and supportive measures, everolimus is generally manageable, but patient education is crucial (e.g., prompt reporting of mouth sores or breathing issues, dietary modifications for diabetes risk, etc.) ⁵⁹ ⁴⁵.

Trial and Guideline Context

Everolimus's incorporation into breast cancer management is supported by clinical trial evidence and reflected in treatment guidelines:

Key Clinical Trials:

- **BOLERO-2 (NCT00863655):** Phase III trial in HR+/HER2– metastatic breast cancer, comparing everolimus + exemestane vs exemestane alone in ~724 postmenopausal women who had progressed on letrozole/anastrozole. Result: **PFS was significantly prolonged** with everolimus (hazard ratio ~0.36–0.43) ⁶³. Median PFS: 6.9 m (everolimus arm) vs 2.8 m (control) by local assessment ⁶³; ~11 m vs 4 m by central review ⁴².

This established the combination as a new standard after AI failure. Overall survival (OS) was later reported not significantly different, though numerically there was a 4.4-month OS advantage favoring everolimus (not meeting significance due to crossover and subsequent therapies) ³⁷. BOLERO-2 also characterized safety (as discussed) and noted consistent PFS benefit across subgroups (including both PIK3CA-mutant and wild-type) ³⁷. This trial led to FDA and EMA approvals in 2012.

- **BOLERO-1 (NCT00876395):** Phase III in **HER2+** first-line metastatic setting (everolimus or placebo + weekly paclitaxel + trastuzumab). It stratified by hormone receptor status. The trial did **not meet its primary endpoint** in the overall population – PFS was ~15 months in both arms ⁶⁴. However, in the pre-planned HR-negative subgroup, there was a notable improvement (20.3 vs 13.1 months, HR 0.66, $p = 0.0049$) ⁶, just shy of the very stringent significance boundary of 0.0044. This suggests everolimus added benefit in HER2+ patients lacking ER. HR-positive patients on this trial might have needed endocrine therapy (which they didn't get) to see benefit – their outcomes were similar with or without everolimus. BOLERO-1's results hint at a targeted niche for everolimus in HER2+ disease (ER-negative cases), but given the overlap of toxicities (some treatment-related deaths occurred in the everolimus arm) ⁴⁵ and the modest effect, it did not change front-line HER2 therapy. It did reinforce biologically that HR status modulates response to mTOR inhibition in HER2+ cancer ⁴⁴.
- **BOLERO-3 (NCT01007942):** Phase III in trastuzumab-resistant **HER2+** metastatic breast cancer (patients previously treated with trastuzumab + chemo). It tested everolimus vs placebo, each added to vinorelbine + trastuzumab. Result: **PFS improved** (median ~7.0 vs 5.8 months, HR 0.78, $p = 0.0067$) in the everolimus arm ³⁶. This was a statistically significant but clinically modest benefit. Notably, an exploratory analysis indicated the benefit was more pronounced in **ER-negative** patients here as well ⁴⁴, consistent with BOLERO-1. BOLERO-3 confirmed that overcoming HER2 resistance via mTOR is viable, though at cost of increased toxicity (stomatitis, cytopenias). It provided the basis for everolimus's inclusion in some guidelines as an option for trastuzumab-resistant cases. However, newer HER2-targeted therapies (T-DM1, pertuzumab, tucatinib, etc.) have largely superseded its use.
- **TAMRAD (UNICANCER):** A smaller Phase II study (everolimus + tamoxifen vs tamoxifen) in metastatic HR+ patients after AI failure. Although phase II, it showed a significant improvement in clinical benefit rate and PFS with the combination. This supported the concept that everolimus can resensitize to hormonal therapy in AI-resistant disease ⁶⁵. TAMRAD hinted that even in patients heavily pretreated, an mTOR inhibitor could restore hormonal responsiveness, thereby influencing the design of BOLERO-2.
- **4EVER, EVEREXT, etc.:** These were single-arm or expanded-access trials evaluating everolimus + exemestane in broader populations or in specific scenarios (like after progression). They generally confirmed the BOLERO-2 findings in the real-world setting: PFS ~6-8 months, safety profile consistent ⁶⁶. The 4EVER trial in particular (phase IIb) in Europe showed a clinical benefit rate ~50% and median PFS ~7.4 months, supporting the effectiveness outside of a strict trial setting ⁶⁶.
- **Brain Mets / TSC Trials:** While not directly breast cancer, it's noteworthy that everolimus (as "Votubia") is used for *tuberous sclerosis complex (TSC)* associated tumors (SEGA, etc.) and has shown CNS penetration and activity. There was exploration whether everolimus could help in breast cancer brain metastases due to PI3K pathway activation there. Some case reports and very small studies suggested modest blood-brain barrier penetration and occasional responses in HER2+ brain mets when adding everolimus to trastuzumab. But no definitive trial.

- **Combination with CDK4/6 or PI3K inhibitors:** Trials are ongoing or recently reported exploring triple combinations, e.g., endocrine + everolimus + ribociclib (CDK4/6i) or everolimus after progression on CDK4/6i. One Phase I (TAKEBACK) suggested everolimus + fulvestrant + ribociclib was feasible and had activity post-CDK4/6. Additionally, to address resistance, trials like BEECH (PI3K+paclitaxel) and others were done; for mTOR specifically, a newer mTORC1/2 inhibitor (vistusertib) in combination with fulvestrant didn't beat everolimus+exemestane in a head-to-head phase II (MANTA trial), highlighting everolimus's benchmark status.

Guidelines (NCCN/ESMO/ASCO/NICE):

- **NCCN (2024):** Recommends everolimus + exemestane as a category 1 option for postmenopausal HR+/HER2- metastatic breast cancer **after progression on a non-steroidal AI**, in patients who do not have a targetable PIK3CA mutation ⁴³. NCCN notes that if a patient has a PIK3CA mutation, alpelisib + fulvestrant is another option; if not or if that fails, everolimus + endocrine is reasonable. NCCN also lists everolimus + fulvestrant (category 2A) as an option after prior CDK4/6 + AI therapy, supported by smaller studies. For HER2+ disease, NCCN mentions that adding everolimus to trastuzumab-based therapy (with chemo) can be considered in later lines (category 2B), referencing BOLERO-3 data, but acknowledges newer agents are preferred. NCCN specifically cautions about pneumonitis and recommends monitoring and prophylactic interventions for stomatitis (e.g., alcohol-free dexamethasone mouthwash).
- **ESMO (2018 Advanced Breast Cancer Guidelines & 2021 update):** ESMO endorses everolimus + exemestane for endocrine-resistant HR+ breast cancer, with a level I, grade A evidence (from BOLERO-2). It suggests this combination after progression on at least one line of hormonal therapy, particularly if CDK4/6 inhibitors are not available or have been used. ESMO also acknowledges everolimus can be combined with fulvestrant (evidence level II). For HER2+, ESMO notes that evidence is insufficient to routinely recommend everolimus, but it could be considered for heavily pre-treated HER2+ patients (especially ER-) if other options are exhausted (expert opinion).
- **ASCO (Endocrine Therapy Guidelines 2021):** ASCO lists everolimus + exemestane as an option after prior endocrine therapy in HR+ MBC, especially after a nonsteroidal AI. They highlight the toxicity and emphasize patient selection and prophylactic mouthwash for stomatitis. ASCO also suggests that in patients progressing on CDK4/6 + AI, everolimus + endocrine can be used, albeit evidence for post-CDK4/6 efficacy is indirect.
- **NICE (UK, 2013 TA295):** Initially, NICE did *not* recommend everolimus + exemestane due to cost-effectiveness concerns. However, Novartis entered a patient access scheme and with updated OS data, NICE reversed the decision, and everolimus has been funded for this indication in the UK for eligible patients (typically those after recurrence on a non-steroidal AI). NICE guidelines mention careful adverse event management. As of the last update, everolimus is recognized as a valid option in England's NHS for HR+ advanced breast cancer after AI failure, in line with its EMA approval, provided the company discount is applied.
- **International Consensus (ABC5 Conference):** Experts agree on the importance of everolimus in HR+ disease, noting it should be considered particularly if *no visceral crisis* (since it's slower-acting than chemo) and if the patient can be closely monitored for side effects. They also note that prophylactic steroid mouthwash has reduced discontinuation rates from stomatitis in practice. For HER2+, the consensus is that better options exist now (e.g., T-DXd, tucatinib), so everolimus is quite down the list.
- **Other Regulatory:** Everolimus is approved by FDA/EMA for HR+ MBC (as above) and by PMDA in Japan (approved 2014) ⁵. The Japanese Breast Cancer Society guidelines mirror Western ones: recommending it in AI-resistant luminal cases.

Current Clinical Context: With the advent of CDK4/6 inhibitors, the sequencing changed: nowadays, first-line for metastatic HR+ is AI + CDK4/6; second-line often fulvestrant + CDK4/6 (if not used) or alpelisib if PIK3CA mutant. Everolimus often comes as a **third-line endocrine-based option** – for example, after AI + CDK4/6 and then fulvestrant ($\pm$ CDK4/6 or alpelisib), one might use everolimus + exemestane or everolimus + fulvestrant. Real-world data (MASCARE registry, etc.) suggests some patients still derive benefit at that point, and it's underused relative to guideline recommendations ⁶⁷. An analysis by Ciruelos et al. 2020 found that everolimus was under-prescribed, possibly due to concern about toxicity, but patients who received it had prolonged OS, implying it contributes meaningfully to disease control ⁶⁷.

For TNBC, everolimus remains investigational. However, given the positive phase II, one could see it being tested in a phase III or incorporated in biomarker-selected cases (e.g., high PI3K pathway signature). Similarly, combinations like everolimus + pembrolizumab in PD-L1+ TNBC are of interest.

Summary: Guidelines place everolimus firmly in HR+ metastatic algorithms (post-AI, evidence Level 1) ⁴³. Its role in other subtypes is smaller: optional in refractory HER2+, not in standard TNBC practice yet. Trials continue to refine this: e.g., CAPItello-290 is testing capivasertib (AKT inhibitor) vs everolimus in HR+ after CDK4/6 – results might further clarify patient selection between different PI3K-axis inhibitors. For now, everolimus is an **integral part of the HR+ MBC armamentarium** and a historical proof-of-concept for targeting tumor escape via the PI3K/mTOR pathway in breast cancer.

Additional Mechanistic or Clinical Notes

- **Mechanism-Based Toxicities and Management:** The adverse effect profile of everolimus (stomatitis, metabolic effects, pneumonitis) is mechanistically tied to mTOR's role in normal tissues. For instance, mTORC1 in oral mucosa basal cells – its inhibition leads to impaired mucosal healing and ulcers. The use of dexamethasone mouthwash prophylaxis (10 mL 4x daily during first 8 weeks) has been shown to reduce grade 2+ stomatitis from ~30% to ~5%, without affecting efficacy. This has become a recommended prophylactic measure in clinical practice, highlighting how understanding the mechanism (local inflammation due to innate immune activation in mouth) led to a preventive strategy. Pneumonitis is thought to result from mTOR inhibition in alveolar cells and immune cells in lung; mechanisms are still studied, but prompt steroid use can reverse it in many cases, indicating an inflammatory component. These mechanistic insights ensure clinicians monitor specific symptoms.
- **Everolimus vs. Alpelisib (PI3K α inhibitor):** In HR+ MBC, alpelisib (a PI3K α -specific inhibitor) is approved for PIK3CA-mutant cases. It might seem to compete with everolimus. Mechanistically, PI3K inhibition (upstream) vs mTORC1 inhibition (downstream) can have different effects: PI3K inhibitors also inhibit AKT activation strongly but may relieve some downstream negative feedback on MAPK; mTORC1 inhibitors relieve AKT feedback but directly shut off protein synthesis. In head-to-head context, SOLAR-1 trial (alpelisib) in PIK3CA-mutants showed PFS ~11 vs 5.7 months with fulvestrant, and overall survival benefit in that subset. Everolimus in BOLERO-2 (mixed mutation status) had PFS ~7 vs 3 months. It's noted that **everolimus works regardless of PIK3CA status** whereas alpelisib requires the mutation for benefit (wild-type PIK3CA had no PFS gain in SOLAR-1). Therefore, in PIK3CA-mutant patients, many guidelines prefer trying alpelisib first (if tolerable, albeit it has its own toxicities like hyperglycemia, rash). Everolimus remains available for those who cannot take alpelisib (e.g., diabetic patients where alpelisib's hyperglycemia is an issue, or those with PIK3CA-wildtype tumors). Some sequencing considerations: If a patient progresses on alpelisib, everolimus can still

be used after, since the mechanism is a bit different and cross-resistance is not complete (though both target the same pathway, resistance to alpelisib might be via isoform switching or downstream mutations, in which case mTOR could still be targetable).

- **Second-generation mTOR inhibitors:** Agents that inhibit both mTORC1 and mTORC2 (dual inhibitors, or “TOR Kinase inhibitors” like vistusertib, sapanisertib) have been explored to overcome resistance (since everolimus doesn’t block mTORC2 and AKT feedback remains an issue). Unfortunately, trials like MANTA (vistusertib + fulvestrant vs everolimus + exemestane) did not show superiority of the dual inhibitor; in fact, everolimus performed better, likely due to dose-limiting toxicities of dual inhibitors and not being able to achieve high exposure. So, everolimus remains the more proven option. There is also interest in combining mTOR inhibitors with others: e.g., **ONC201 (Dopamine receptor antagonist)** was tested to enhance everolimus’s durability (by causing dual inhibition of mTOR and activating an integrated stress response) ²⁵; early results suggested some synergistic effect in ER+ models, but clinical translation is in early phases.
- **Immunotherapy Combinations:** Given the immunosuppressive effects of mTOR inhibitors, one might think they conflict with immunotherapy. Paradoxically, the **preclinical TNBC study (Li et al., Cell Signal 2023)** ⁴⁷ ³⁵ showed that combining everolimus with anti-PD-1 therapy produced better tumor control than either alone – everolimus normalized tumor vasculature and reduced regulatory immune elements when PD-1 was blocked, leading to improved CD8 T-cell infiltration and tumor regression. This suggests a potentially promising strategy: use everolimus to starve tumors and modulate the microenvironment, and use checkpoint inhibitors to reinvigorate T-cells. A clinical trial could explore everolimus + atezolizumab or pembrolizumab in TNBC (particularly if the TNBC has PI3K pathway activation and PD-L1 positivity). This is a cutting-edge idea bridging targeted therapy and immunotherapy.
- **Liquid Biopsy for Resistance Monitoring:** As patients undergo everolimus therapy, tracking emerging resistance via blood biomarkers is an area of research. For example, rising levels of mutant PIK3CA circulating DNA, or emergence of MTOR mutations (some MTOR mutations in the FKBP12 binding domain can cause rapalog resistance) might be detectable. Also, circulating exosomes carrying phosphorylated proteins from the tumor (p-AKT, etc.) might indicate pathway reactivation. While not in clinical use, these approaches are being studied to predict when a patient is losing benefit from everolimus and should switch therapy.
- **Special situations:** Everolimus has shown efficacy in some hormone-positive **male** breast cancer cases (though data are very limited, male MBC can be treated similarly to female HR+ MBC with everolimus + exemestane if needed). Also, for patients with **organ transplants** who develop breast cancer, everolimus could theoretically kill two birds – treat cancer and maintain graft – since it’s used as an immunosuppressant. Case reports have described using everolimus in a patient with a renal transplant and breast cancer, successfully controlling both transplant rejection and cancer, which is a unique scenario where its dual role is beneficial (though careful balancing of immunosuppression vs. tumor control was needed).
- **Quality of Life and adherence:** Everolimus can impact QoL with its chronic toxicities. Stomatitis especially can reduce oral intake and weight. Early and aggressive supportive care is vital to keep patients on drug long enough to benefit (PFS ~6-7 months means many months of therapy). Data suggests patients who required dose reductions often still derived benefit, so tolerability-guided

dose adjustment (to 5 mg daily, or even 5 mg every-other-day in frail patients) is an option. Patient education to report side effects early leads to interventions that prevent treatment discontinuation. Real-world studies show that in practices that are experienced with mTORi, discontinuation rates due to toxicity drop significantly.

- **Everolimus in Early Breast Cancer:** One might wonder if mTOR inhibitors have a role in high-risk early breast cancer (adjuvant setting). So far, trials like UNIRAD (everolimus + adjuvant endocrine vs endocrine alone in high-risk ER+ after chemo) have not reported a positive result (the UNIRAD trial did not show a significant invasive disease-free survival benefit, presented 2020). The toxicity in the adjuvant setting was a concern and may have outweighed potential benefit. Thus, at present, everolimus is confined to metastatic disease.
- **Pharmacogenomics:** No strong pharmacogenomic predictors of everolimus metabolism or toxicity are established. Some data indicates that patients with certain CYP3A5 genotypes or ABCB1 (P-glycoprotein) polymorphisms may have different clearance, but nothing that requires testing before use. Therapeutic drug monitoring is standard in transplant, not in oncology – but some argue checking everolimus trough levels (target ~10 ng/mL) could help manage toxicity vs efficacy in outlier cases.

In closing, **everolimus** brought a new mechanism into breast oncology – targeting tumor cell metabolism and growth signals – and paved the way for subsequent drugs targeting the PI3K axis. It remains a valuable option, particularly in **endocrine therapy-resistant luminal breast cancer**, and ongoing research continues to refine its use, mitigate its toxicities, and expand its benefits via rational combinations.

Pathway Evidence Table

Pathway (ID/Name)	Regulation	Effect of Everolimus	Biological Rationale	Reference(s)
HALLMARK_PI3K_AKT_MTOR_SIGNALING	Down	Sensitive (target)	Core oncogenic pathway inhibited by everolimus. Hyperactivation of PI3K/AKT (e.g. by PIK3CA mutation or PTEN loss) drives proliferation and endocrine resistance 4 9 . Everolimus blocks mTORC1, shutting down this pathway and causing tumor growth arrest. Tumors addicted to PI3K/mTOR are highly sensitive, whereas reactivation of AKT signaling confers resistance if this pathway remains up 25 .	4 9 25

Pathway (ID/Name)	Regulation	Effect of Everolimus	Biological Rationale	Reference(s)
HALLMARK_MTORC1_SIGNALING	Down	Sensitive (target)	Everolimus directly inhibits mTORC1 activity, reducing phosphorylation of S6K and 4EBP1 ¹⁸ . This halts cap-dependent translation and cell-cycle progression (reflected by reduced Cyclin D1, c-Myc levels) ²⁶ . Downregulation of mTORC1's output is essential for anti-tumor effect (sensitivity). Tumors that cannot sufficiently suppress mTORC1 signaling (e.g. due to mTOR activation mutations) exhibit resistance.	¹⁸ ²⁶

Pathway (ID/Name)	Regulation	Effect of Everolimus	Biological Rationale	Reference(s)
HALLMARK_ESTROGEN_RESPONSE (ER signaling)	Down	Sensitive (contextual)	Everolimus downregulates ERα expression and transcriptional activity in HR+ breast cancer cells ²⁰ . This mitigates estrogen-driven growth and helps overcome endocrine resistance. In ER+ tumors with active ER signaling, everolimus enhances sensitivity by blocking PI3K/ mTOR cross-talk that sustains ER function ²⁷ . If ER pathway is lost (downregulated), the combination of everolimus + endocrine is less effective (resistance in HR- tumors) ⁴⁴ .	²⁰ ²⁷ > ⁴⁴

Pathway (ID/Name)	Regulation	Effect of Everolimus	Biological Rationale	Reference(s)
HALLMARK_E2F_TARGETS (Cell cycle)	Down	Sensitive (mechanism)	Inhibition of mTORC1 causes G ₁ arrest, reflected by downregulation of E2F-driven genes (Cyclin E, etc.) ²⁶ . Everolimus-treated cells accumulate in G ₁ due to lower Cyclin D1 and p21 levels ²¹ . Tumors with high proliferation rates (high E2F targets) depend on mTORC1 for protein synthesis; their cell-cycle is more drastically halted (sensitive). If alternate cell-cycle drivers (e.g. CDK2 activation via other pathways) are up, cells may resist despite E2F downregulation.	²⁶

Pathway (ID/Name)	Regulation	Effect of Everolimus	Biological Rationale	Reference(s)
Autophagy (GO:0006914)	Up	Resistant	mTORC1 normally suppresses autophagy; everolimus <i>induces autophagy</i> as an adaptive survival route ²⁸. Breast cancer cells on everolimus show increased LC3-II and autophagosome formation (autophagy up) ²⁸. This pathway's upregulation lets tumor cells recycle nutrients and avoid apoptosis, leading to drug resistance (everolimus insensitivity). Inhibiting autophagy reverses this resistance in preclinical models, confirming autophagy-up is a resistance mechanism.	²⁸

Pathway (ID/Name)	Regulation	Effect of Everolimus	Biological Rationale	Reference(s)
MYC_Oncogenic_Signaling (e.g. HALLMARK_MYC_TARGETS)	Up	Resistant	Everolimus resistance is driven by <i>upregulation of c-MYC</i> oncogene and its transcriptional program ²⁹ . Resistant ER+ cells show elevated MYC mRNA/protein and enriched MYC target gene signatures ²⁹ . MYC upregulation sustains growth and metabolism independently of mTORC1, bypassing the drug's effect. Functionally, knocking down MYC resensitizes cells to everolimus (proving MYC-up causes resistance) ³⁰ . Clinically, tumors with high MYC expression or amplification respond poorly to everolimus.	²⁹ ³⁰

Pathway (ID/Name)	Regulation	Effect of Everolimus	Biological Rationale	Reference(s)
AKT & Receptor Tyrosine Kinase Signaling (e.g. REACTOME_PI3K_AKT, HER2/IGF1R)	Up	Resistant	mTORC1 inhibition relieves feedback inhibition on insulin/IGF receptors and HER receptors, often causing <i>AKT hyper-activation</i> as a compensatory response ³² . Upregulation of upstream RTKs (IGF1R, EGFR, HER2/HER3) and increased AKT phosphorylation have been observed at progression on everolimus ²⁵ . This pathway up (PI3K/ AKT reactivation) allows the tumor to maintain survival signaling (via mTORC2 or alternate routes) despite everolimus, leading to treatment resistance. In HER2+ tumors, increased HER3–PI3K signaling correlates with everolimus non-response, highlighting RTK/ AKT upregulation as a resistance driver.	²⁵ ³²

Pathway (ID/Name)	Regulation	Effect of Everolimus	Biological Rationale	Reference(s)
STAT3-HIF-VEGF Angiogenesis	Down	Sensitive (mechanism)	Everolimus inhibits tumor angiogenesis by <i>downregulating the STAT3 → HIF-1α → VEGF pathway</i>. In TP53-mutant solid tumors, everolimus significantly reduced HIF-1α and VEGF levels, decreasing microvessel density ²² ²³. This translates to nutrient deprivation and growth suppression of tumor tissue (sensitive effect). If a tumor finds alternative angiogenic pathways or if HIF-1α is reactivated (e.g., via MAPK or P53 loss independent of mTOR), resistance can occur. Generally, VEGF pathway downregulation is part of everolimus's anti-tumor action.	²² ²³

Pathway (ID/Name)	Regulation	Effect of Everolimus	Biological Rationale	Reference(s)
Immune Checkpoint/PD-L1 Pathway	Up	Resistant	Everolimus's immunosuppressive milieu can lead to <i>upregulation of PD-L1 and other immune evasion signals</i>, blunting anti-tumor immunity ³⁴ ³⁵ . In TNBC models, everolimus increased PD-L1 on endothelial cells and reduced CD8⁺ T-cell tumor infiltration, allowing tumors to progress unless PD-1 blockade was added ³⁴ ³⁵ . Thus, an upregulated PD-1/ PD-L1 pathway (or generally, heightened immune checkpoint activity) under everolimus is associated with resistance. Combining checkpoint inhibitors can counteract this; otherwise, tumors with an "immune-cold" microenvironment on everolimus are less likely to respond.	³⁴ ³⁵

ER (Estrogen Receptor) Pathway

Down

Resistant

Loss or downregulation of ER signaling in formerly ER+ disease confers resistance to everolimus + endocrine therapy. Everolimus requires concurrent hormone blockade to be effective in ER+ cancer ⁴. If a tumor downregulates ER (through ESR1 mutation causing ligand-independent growth, or by completely losing ER expression), it no longer relies on estrogen and often activates alternate pathways (like MAPK or AKT) unaffected by everolimus. Clinical evidence: HER2+ patients with ER-positive tumors derived no benefit from everolimus without endocrine therapy, whereas ER-negative did ⁶⁴⁴. This suggests ER pathway down = everolimus resistance in that context. Therefore, the absence of ER-driven signaling (a downregulated ER pathway) is a resistance indicator to mTOR-targeted

⁶ ⁴⁴

> ⁴

Pathway (ID/Name)	Regulation	Effect of Everolimus	Biological Rationale	Reference(s)
			augmentation of endocrine therapy.	

Notes: “Upregulated” or “Downregulated” refers to the pathway’s activity/expression in the tumor context with everolimus. “Effect of Everolimus” denotes whether the pathway’s status contributes to sensitivity or resistance to the drug. For example, upregulation of a bypass pathway (like MYC or AKT) leads to drug resistance, whereas upregulation of the target pathway (PI3K/mTOR) indicates sensitivity (due to dependency) ⁴ ⁹. All evidence is from human studies or human-derived models in breast cancer. The references support each pathway’s inclusion and role, drawn from clinical trials, translational research, and curated databases as detailed above.

¹ ¹¹ ¹⁶ Everolimus: Uses, Interactions, Mechanism of Action | DrugBank Online
<https://go.drugbank.com/drugs/DB01590>

² ¹² ¹³ ¹⁵ ³³ Definition of everolimus - NCI Drug Dictionary - NCI
<https://www.cancer.gov/publications/dictionaries/cancer-drug/def/everolimus>

³ ⁴¹ ⁴² ⁵⁹ ⁶³ Everolimus in postmenopausal hormone-receptor-positive advanced breast cancer - PubMed
<https://pubmed.ncbi.nlm.nih.gov/22149876/>

⁴ ⁵ ²⁴ ³⁷ Overcoming the Adverse Effects of Everolimus to Achieve Maximum Efficacy in the Treatment of Inoperable Breast Cancer: A Review of 11 Cases at Our Hospital | Case Reports in Oncology | Karger Publishers
<https://karger.com/cro/article/11/2/511/90200/Overcoming-the-Adverse-Effects-of-Everolimus-to>

⁶ ³⁶ ⁴⁴ ⁴⁵ ⁶⁰ ⁶¹ ⁶⁴ BOLERO-1: Everolimus Plus Trastuzumab/Paclitaxel Misses the Mark in First-Line HER2-Positive Advanced Breast Cancer - The ASCO Post
<https://ascopost.com/issues/january-25-2015/bolero-1-everolimus-plus-trastuzumabpaclitaxel-misses-the-mark-in-first-line-her2-positive-advanced-breast-cancer/>

⁷ ⁸ ⁹ ⁴⁶ Added Everolimus Reduces Risk for Advanced TNBC - The ASCO Post
<https://ascopost.com/news/august-2025/added-everolimus-reduces-risk-for-advanced-tnbc/>

¹⁰ Compound: EVEROLIMUS (ChEMBL1908360) - ChEMBL - EMBL-EBI
<https://www.ebi.ac.uk/chembl/explore/compound/ChEMBL1908360>

¹⁴ ¹⁷ New Drug Approvals - Pt. V - Everolimus (Afinitor)
<http://chembl.github.io/new-drug-approvals-pt-v-everolimus/>

¹⁸ ¹⁹ ²⁰ ²¹ ²⁶ ²⁷ ²⁸ ⁶⁵ Everolimus downregulates estrogen receptor and induces autophagy in aromatase inhibitor-resistant breast cancer cells | BMC Cancer | Full Text
<https://bmccancer.biomedcentral.com/articles/10.1186/s12885-016-2490-z>

²² ²³ Everolimus downregulates STAT3/HIF-1 α /VEGF pathway to inhibit angiogenesis and lymphangiogenesis in TP53 mutant head and neck squamous cell... | Oncotarget
<https://www.oncotarget.com/article/28355/text/>

25 32 **ONC201/TIC10 enhances durability of mTOR inhibitor everolimus in ...**

<https://elifesciences.org/articles/85898>

29 30 31 **Resistance to everolimus driven by epigenetic regulation of MYC in ER+ breast cancers | Oncotarget**

<https://www.oncotarget.com/article/2964/text/>

34 35 47 **Everolimus combined with PD-1 blockade inhibits progression of triple-negative breast cancer - PubMed**

<https://pubmed.ncbi.nlm.nih.gov/37257766/>

38 **Resistance to the mTOR inhibitor everolimus is reversed by the ...**

<https://www.spandidos-publications.com/10.3892/ol.2017.6597>

39 40 **prompt1_updated_drug_effect.docx**

file:///file_000000000f647206b3cb5791fed11bde

43 **Expert consensus on treating HR+/HER2- metastatic breast cancer ...**

<https://pmc.ncbi.nlm.nih.gov/articles/PMC12138406/>

48 **DailyMed - EVEROLIMUS tablet**

<https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=59cb86d5-8706-4619-b074-af253d3e0b68>

49 50 51 52 53 54 55 56 57 58 62 **AFINITOR (everolimus) tablets**

https://www.accessdata.fda.gov/drugsatfda_docs/label/2012/022334s016lbl.pdf

66 **overall survival results from BOLERO-2 - Annals of Oncology**

[https://www.annalsofoncology.org/article/S0923-7534\(19\)36905-4/fulltext](https://www.annalsofoncology.org/article/S0923-7534(19)36905-4/fulltext)

67 **Long-Term Results with Everolimus in Advanced Hormone Receptor ...**

<https://www.mdpi.com/2072-6694/15/4/1191>